

Customer Churn Prediction – Full Report

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1. Problem statement

Predict which subscription customers are likely to churn, identify key drivers of churn and support targeted retention.

2. Methodology

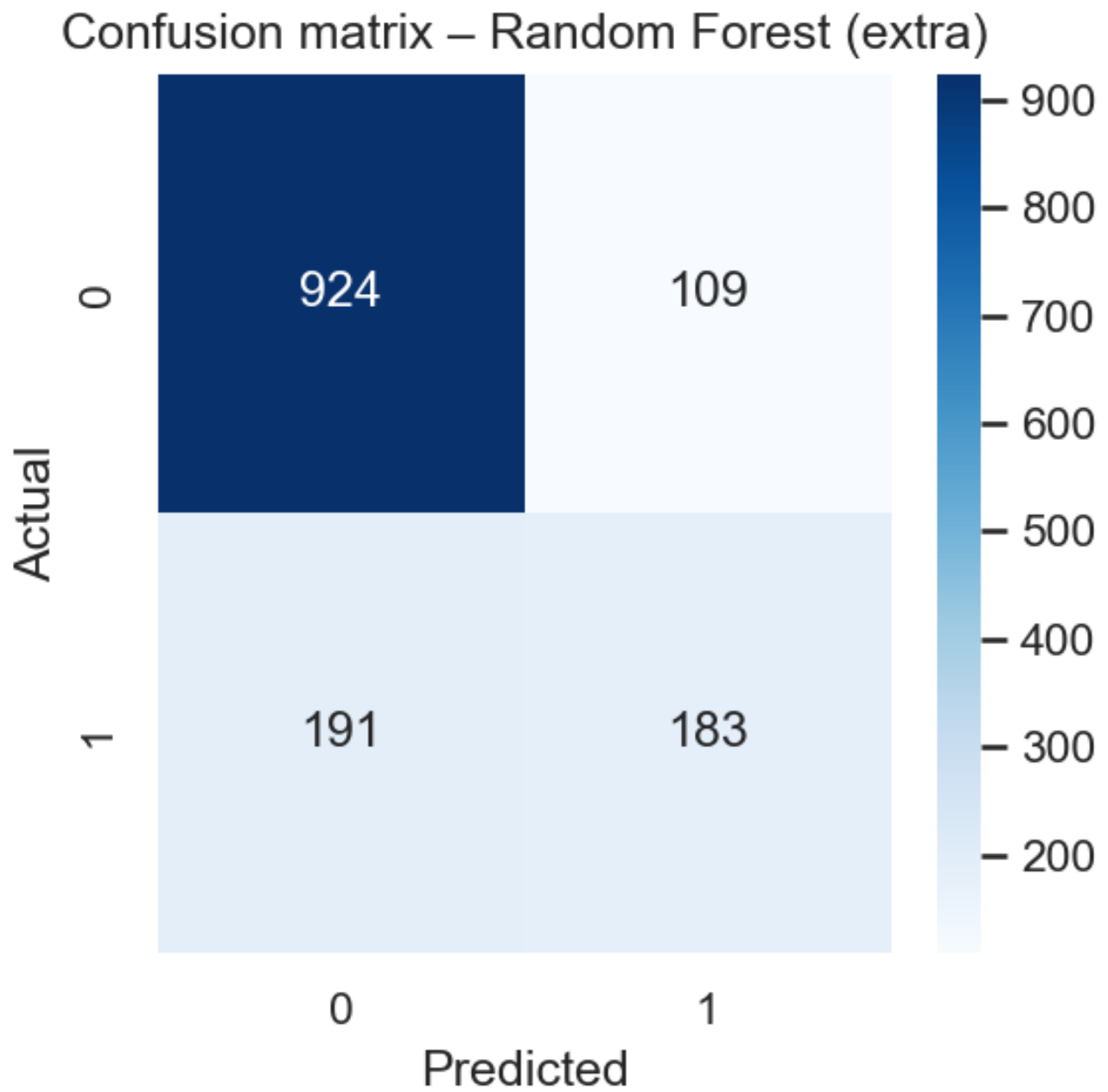
- EDA on churn distribution and key drivers (tenure, charges, contract type).
- Data cleaning, encoding of categorical variables, scaling of numeric variables.
- Training and comparison of Logistic Regression and Random Forest models.
- Evaluation with accuracy, precision, recall, F1 and ROC AUC on a test set.

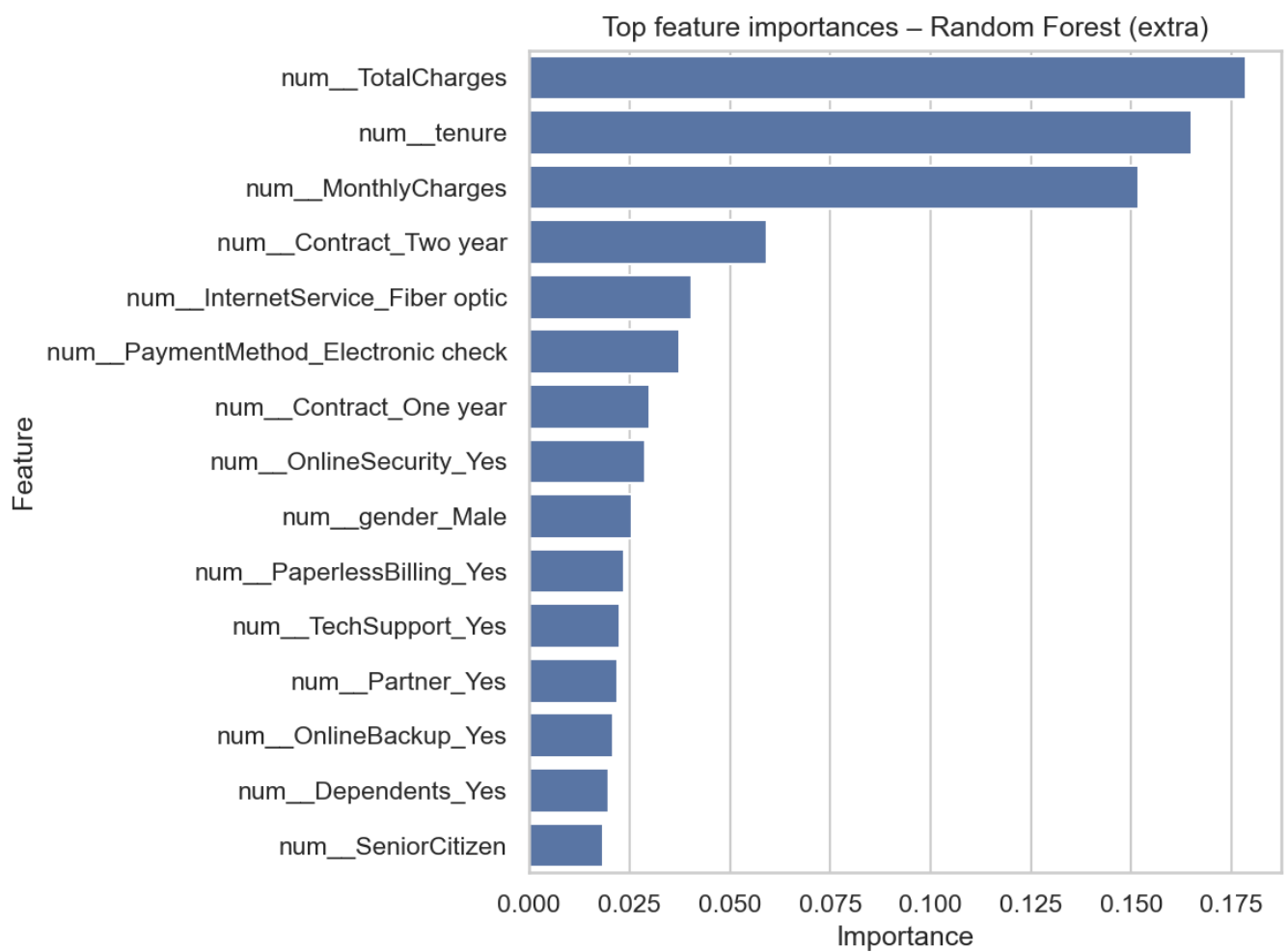
3. Model performance summary

Below is the visual model comparison table from the notebook.

model_summary_extra.png

model		accuracy	precision	recall	f1	roc_auc
Logistic Regression (extra)	Logistic Regression (extra)	0.804	0.648	0.575	0.609	0.836
	Random Forest (extra)	0.787	0.627	0.489	0.55	0.819





4. Recommendations

- Use the best performing model to score customers regularly and focus on high risk deciles.
- Prioritise high value customers with high churn scores for proactive retention actions.
- Review onboarding and communication for short tenure customers and month to month contracts.
- Consider further model tuning and adding behavioural features such as usage and support tickets.